

ABOUT THIS BOOK

This book introduces matrices through high school coordinate geometry. This approach makes it easier for beginners to learn Python for scientific computing. All problems in the book are from NCERT mathematics textbooks from Class 9-12. Exercises are from CBSE board exam papers.

The content is sufficient for industry jobs and covers nearly all matrix prerequisites for machine learning. There is no copyright, so readers are free to print and share.

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In this edition, some incorrect solutions were removed. All figures redrawn. More problems added.

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1. GATE CE-2 2025

A. Formulae

1.1.1 Exponential Series

- **Taylor and Maclaurin Series Foundations:** Let $f(x)$ be an infinitely differentiable function in a neighborhood of $x = x_0$. The Taylor series expansion of $f(x)$ centered at x_0 is defined as:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(x_0)}{n!} (x - x_0)^n \quad (1.1.1.1)$$

where $f^{(n)}(x_0)$ denotes the n -th order derivative evaluated at $x = x_0$, with $f^{(0)}(x_0) = f(x_0)$ and $0! = 1$.

When the series is centered specifically about the origin ($x_0 = 0$), it is referred to as the Maclaurin series:

$$f(x) = \sum_{n=0}^{\infty} \frac{f^{(n)}(0)}{n!} x^n = f(0) + \frac{f'(0)}{1!} x + \frac{f''(0)}{2!} x^2 + \frac{f'''(0)}{3!} x^3 + \dots \quad (1.1.1.2)$$

- **Derivation of the Exponential Expansion:** Consider the exponential function $f(x) = e^x$. The successive derivatives of e^x with respect to x are invariant:

$$f'(x) = \frac{d}{dx} (e^x) = e^x \quad (1.1.1.3)$$

$$f''(x) = \frac{d^2}{dx^2} (e^x) = e^x \quad (1.1.1.4)$$

$$\vdots$$

$$f^{(n)}(x) = \frac{d^n}{dx^n} (e^x) = e^x, \quad n \geq 0 \quad (1.1.1.5)$$

Evaluating each derivative at $x = 0$:

$$f^{(n)}(0) = e^0 = 1, \quad n \geq 0 \quad (1.1.1.6)$$

Substituting $f^{(n)}(0) = 1$ directly into the Maclaurin series formulation (1.1.1.2) yields the power series expansion for e^x :

$$e^x = \sum_{n=0}^{\infty} \frac{1}{n!} x^n = 1 + \frac{x}{1!} + \frac{x^2}{2!} + \frac{x^3}{3!} + \dots \quad (1.1.1.7)$$

- **Absolute Convergence via Ratio Test:** Let $a_n = \frac{x^n}{n!}$. The interval of convergence is determined using the ratio test:

$$L = \lim_{n \rightarrow \infty} \left| \frac{a_{n+1}}{a_n} \right| = \lim_{n \rightarrow \infty} \left| \frac{x^{n+1}}{(n+1)!} \cdot \frac{n!}{x^n} \right| = \lim_{n \rightarrow \infty} \frac{|x|}{n+1} = 0 \quad (1.1.1.8)$$

Since $L = 0 < 1$ for all finite $x \in \mathbb{R}$, the series converges absolutely on $(-\infty, \infty)$ with an infinite radius of convergence ($R = \infty$).

1.1.2 Intersection of Conics and Common Chord

- **Line-Conic Intersection:** The points of intersection of a line $L : \mathbf{x} = \mathbf{h} + \kappa \mathbf{m}$ ($\kappa \in \mathbb{R}$) with a conic section $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$ are given by:

$$\mathbf{x}_i = \mathbf{h} + \kappa_i \mathbf{m} \quad (1.1.2.1)$$

where the parameter values κ_i are:

$$\kappa_i = \frac{-\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u}) \pm \sqrt{[\mathbf{m}^\top (\mathbf{V} \mathbf{h} + \mathbf{u})]^2 - g(\mathbf{h}) (\mathbf{m}^\top \mathbf{V} \mathbf{m})}}{\mathbf{m}^\top \mathbf{V} \mathbf{m}} \quad (1.1.2.2)$$

with $g(\mathbf{h}) = \mathbf{h}^\top \mathbf{V} \mathbf{h} + 2\mathbf{u}^\top \mathbf{h} + f$.

- **Matrix Formulation of Intersection Points:** The intersection coordinates can be written in matrix form as:

$$\mathbf{X} = (\mathbf{h} \quad \mathbf{m}) \begin{pmatrix} 1 & 1 \\ \kappa_1 & \kappa_2 \end{pmatrix} \quad (1.1.2.3)$$

- **Degenerate Conic Condition:** A quadratic form $\mathbf{x}^\top \mathbf{V} \mathbf{x} + 2\mathbf{u}^\top \mathbf{x} + f = 0$ represents a pair of straight lines if and only if the augmented matrix is singular:

$$\begin{vmatrix} \mathbf{V} & \mathbf{u} \\ \mathbf{u}^\top & f \end{vmatrix} = 0 \quad (1.1.2.4)$$

- **Pencil of Conics and Common Chord:** The intersection of two conics with parameters $(\mathbf{V}_1, \mathbf{u}_1, f_1)$ and $(\mathbf{V}_2, \mathbf{u}_2, f_2)$ is given by the pencil:

$$\mathbf{x}^\top (\mathbf{V}_1 + \mu \mathbf{V}_2) \mathbf{x} + 2(\mathbf{u}_1 + \mu \mathbf{u}_2)^\top \mathbf{x} + (f_1 + \mu f_2) = 0 \quad (1.1.2.5)$$

The value of μ corresponding to the degenerate conic (the radical axis / common chord line) is found from:

$$\begin{vmatrix} \mathbf{V}_1 + \mu \mathbf{V}_2 & \mathbf{u}_1 + \mu \mathbf{u}_2 \\ (\mathbf{u}_1 + \mu \mathbf{u}_2)^\top & f_1 + \mu f_2 \end{vmatrix} = 0 \quad (1.1.2.6)$$

1.1.3 Laplace Transform

- **Bilateral Definition:** For an arbitrary continuous-time signal $g(t)$, the bilateral Laplace transform is defined as:

$$G(s) = \mathcal{L}\{g(t)\} = \int_{-\infty}^{\infty} g(t) e^{-st} dt \quad (1.1.3.1)$$

- **Unilateral Definition:** For causal signals and differential equations with initial conditions at $t = 0$:

$$G(s) = \mathcal{L}\{g(t)\} = \int_0^{\infty} g(t) e^{-st} dt \quad (1.1.3.2)$$

- **Derivation of Derivative Properties:** Applying integration by parts to $\mathcal{L}\{y'(t)\}$

with $u = e^{-st}$ and $dv = y'(t) dt \implies du = -se^{-st} dt$, $v = y(t)$:

$$\mathcal{L}\{y'(t)\} = \int_0^\infty y'(t) e^{-st} dt \quad (1.1.3.3)$$

$$= [y(t) e^{-st}]_0^\infty - \int_0^\infty y(t) (-se^{-st}) dt \quad (1.1.3.4)$$

$$= (0 - y(0)) + s \int_0^\infty y(t) e^{-st} dt \quad (1.1.3.5)$$

$$= sY(s) - y(0) \quad (1.1.3.6)$$

Thus, the first derivative transform is:

$$y'(t) \xleftrightarrow{\mathcal{L}} sY(s) - y(0) \quad (1.1.3.7)$$

Applying the first derivative property recursively to the second derivative $y''(t) = \frac{d}{dt}(y'(t))$:

$$\mathcal{L}\{y''(t)\} = s\mathcal{L}\{y'(t)\} - y'(0) \quad (1.1.3.8)$$

$$= s[sY(s) - y(0)] - y'(0) \quad (1.1.3.9)$$

$$= s^2Y(s) - sy(0) - y'(0) \quad (1.1.3.10)$$

Under zero initial conditions ($y(0) = 0$, $y'(0) = 0$), the operational transforms simplify to:

$$y(t) \xleftrightarrow{\mathcal{L}} Y(s) \quad (1.1.3.11)$$

$$y'(t) \xleftrightarrow{\mathcal{L}} sY(s) \quad (1.1.3.12)$$

$$y''(t) \xleftrightarrow{\mathcal{L}} s^2Y(s) \quad (1.1.3.13)$$

- **Transform of the Unit Step Function:** The Laplace transform of an unconstrained constant $f(t) = 1$ over $(-\infty, \infty)$ is undefined because the integral diverges as $t \rightarrow -\infty$. For causal systems, a constant input starting at $t = 0$ is defined through the unit step function $u(t)$:

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases} \quad (1.1.3.14)$$

Evaluating the unilateral Laplace transform of $u(t)$:

$$\mathcal{L}\{u(t)\} = \int_0^\infty 1 \cdot e^{-st} dt = \left[-\frac{e^{-st}}{s} \right]_0^\infty = \frac{1}{s}, \quad \text{Re}(s) > 0 \quad (1.1.3.15)$$

Hence:

$$u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s} \quad (1.1.3.16)$$

- Transform of e^{at} :

$$e^{at}u(t) \xleftrightarrow{\mathcal{L}} \frac{1}{s-a}, \quad \operatorname{Re}(s) > a \quad (1.1.3.17)$$

$$(1.1.3.18)$$

1.1.4 Euler's Method of Forward Differences

- For a first-order differential equation $\frac{dy}{dt} = f(t, y)$ with step size h :

$$y_{n+1} = y_n + hf(t_n, y_n) \quad (1.1.4.1)$$

1.1.5 Gradient Descent Optimization

- First-Order Parameter Update: For a differentiable function $f(x)$, the iterative parameter update with learning rate $\eta > 0$ is given by:

$$x_{k+1} = x_k - \eta f'(x_k) \quad (1.1.5.1)$$

- Second Derivative Test for Convexity:

$$f''(x) > 0 \quad \forall x \in (0, \infty) \implies f(x) \text{ is strictly convex} \quad (1.1.5.2)$$

A strictly convex function on an open interval possesses at most one local minimum, which is also its unique global minimum.

1.1.6 Minimization via Completing the Square

- For any $x > 0$ and constant $a > 0$, expanding the squared difference gives:

$$\left(\sqrt{x} - \frac{a}{\sqrt{x}}\right)^2 = x - 2a + \frac{a^2}{x} \quad (1.1.6.1)$$

Because the square of any real term is non-negative $\left(\sqrt{x} - \frac{a}{\sqrt{x}}\right)^2 \geq 0$, rearranging yields the lower bound:

$$x + \frac{a^2}{x} = \left(\sqrt{x} - \frac{a}{\sqrt{x}}\right)^2 + 2a \geq 2a \quad (1.1.6.2)$$

The equality holds if and only if the squared term vanishes:

$$\sqrt{x} - \frac{a}{\sqrt{x}} = 0 \implies x = a \quad (1.1.6.3)$$

attaining the unique minimum value $2a$.

1.1.7 Newton-Raphson Method

- First-Order Recurrence (Difference Equation): For a real-valued differentiable function $f(x)$, the root is iteratively approximated using its first-order Taylor series expansion about the current estimate x_n :

$$x_{n+1} = x_n - \frac{f(x_n)}{f'(x_n)} \quad (1.1.7.1)$$

1.1.8 System of Linear Equations and Rouché–Capelli Theorem

- A system of m linear equations in n variables can be represented in matrix form

as:

$$\mathbf{A}\mathbf{x} = \mathbf{b} \quad (1.1.8.1)$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is the coefficient matrix, $\mathbf{x} \in \mathbb{R}^{n \times 1}$ is the vector of unknowns, and $\mathbf{b} \in \mathbb{R}^{m \times 1}$ is the constant vector.

- The solvability of the system is governed by the ranks of the coefficient matrix $\mathbf{A}$ and the augmented matrix $[\mathbf{A} \mid \mathbf{b}]$:
 - a) Inconsistent System (No Solution): A linear system has no solution if and only if:

$$\text{rank}(\mathbf{A}) < \text{rank}([\mathbf{A} \mid \mathbf{b}]) \quad (1.1.8.2)$$

Geometrically, this corresponds to distinct parallel lines that do not intersect.

- b) Consistent System (At Least One Solution): A linear system has at least one solution if and only if:

$$\text{rank}(\mathbf{A}) = \text{rank}([\mathbf{A} \mid \mathbf{b}]) = r \quad (1.1.8.3)$$

- If $r = n$ (where n is the number of variables), there exists a unique solution.
- If $r < n$, there exist infinitely many solutions with $n - r$ free variables (coincident lines).

1.1.9 Sum of Independent Discrete Random Variables (Two Dice Roll)

- Let $X_1, X_2 \in \{1, 2, 3, 4, 5, 6\}$ be independent discrete uniform random variables representing the outcomes of two fair dice, each having probability mass function (PMF):

$$p_{X_1}(n) = p_{X_2}(n) = \begin{cases} \frac{1}{6}, & 1 \leq n \leq 6 \\ 0, & \text{otherwise} \end{cases} \quad (1.1.9.1)$$

- Let $X = X_1 + X_2$ denote their sum. The PMF of X is given by the discrete convolution of $p_{X_1}(n)$ and $p_{X_2}(n)$:

$$p_X(n) = \Pr(X_1 + X_2 = n) = \sum_k \Pr(X_1 = n - k \mid X_2 = k) p_{X_2}(k) \quad (1.1.9.2)$$

$$= \sum_k p_{X_1}(n - k) p_{X_2}(k) = p_{X_1}(n) * p_{X_2}(n) \quad (1.1.9.3)$$

- Substituting (1.1.9.1) into (1.1.9.3) yields the triangular PMF:

$$p_X(n) = \begin{cases} 0, & n < 2 \\ \frac{n-1}{36}, & 2 \leq n \leq 7 \\ \frac{13-n}{36}, & 7 < n \leq 12 \\ 0, & n > 12 \end{cases} \quad (1.1.9.4)$$

1.1.10 Unit Step Response of an RC Circuit and s -Domain Analysis

- Time-Domain Circuit Equation: Consider a series RC circuit excited by a step

input $v_i(t) = V_{dc}u(t)$ where V_{dc} is a constant DC voltage source, and $u(t)$ is step input function defined as:

$$u(t) = \begin{cases} 1, & t \geq 0 \\ 0, & t < 0 \end{cases} \quad (1.1.10.1)$$

Starting at $t = 0$, the current through the capacitor is:

$$i(t) = \frac{dq}{dt} = C \frac{dV_{out}}{dt} \quad (1.1.10.2)$$

Applying Kirchhoff's Voltage Law (KVL) around the loop:

$$Ri(t) + V_{out}(t) = V_{dc} \implies RC \frac{dV_{out}}{dt} + V_{out}(t) = V_{dc} \quad (1.1.10.3)$$

- *s*-Domain Representation: Transforming the circuit to the Laplace domain, the step source becomes $V_{in}(s) = \frac{V_{dc}}{s}$, the resistor impedance is R , and the capacitor impedance is $\frac{1}{sC}$. The equivalent impedance is:

$$R_{eq}(s) = R + \frac{1}{sC} = \frac{RsC + 1}{sC} \quad (1.1.10.4)$$

The loop current in the *s*-domain is:

$$I(s) = \frac{V_{in}(s)}{R_{eq}(s)} = \frac{\frac{V_{dc}}{s}}{\frac{RsC + 1}{sC}} = \frac{V_{dc}C}{RsC + 1} \quad (1.1.10.5)$$

The output voltage across the capacitor is obtained as:

$$V_{out}(s) = I(s) \left(\frac{1}{sC} \right) = \frac{V_{dc}}{s(RCs + 1)} \quad (1.1.10.6)$$

$$= \frac{V_{dc}}{RC} \left[\frac{1}{s(s + \frac{1}{RC})} \right] = V_{dc} \left[\frac{1}{s} - \frac{1}{s + \frac{1}{RC}} \right] \quad (1.1.10.7)$$

Taking the inverse Laplace transform yields:

$$V_{out}(t) = V_{dc} \left(1 - e^{-\frac{t}{RC}} \right) u(t) \quad (1.1.10.8)$$

1.1.11 Standard First-Order Differential Equation Comparison

- A general first-order dynamic system with time constant τ subjected to a unit step input is described by:

$$\tau \frac{dy}{dt} + y(t) = y_{ss} \quad (1.1.11.1)$$

where τ is the time constant and y_{ss} is the steady-state response.

- Comparing (1.1.11.1) with the RC circuit in (1.1.10.3):

$$y(t) \longleftrightarrow V_{out}(t), \quad \tau \longleftrightarrow RC, \quad y_{ss} \longleftrightarrow V_{dc} \quad (1.1.11.2)$$

Transforming (1.1.11.1) into the s -domain with zero initial conditions $y(0) = 0$:

$$\tau s Y(s) + Y(s) = \frac{y_{ss}}{s} \implies Y(s) = \frac{y_{ss}}{s(\tau s + 1)} = y_{ss} \left[\frac{1}{s} - \frac{1}{s + \frac{1}{\tau}} \right] \quad (1.1.11.3)$$

Taking the inverse Laplace transform directly gives:

$$y(t) = y_{ss} \left(1 - e^{-\frac{t}{\tau}} \right) u(t) \quad (1.1.11.4)$$

1.1.12 Transfer Function of a First-Order Process

- In continuous-time signals and linear time-invariant (LTI) systems, the transfer function $H(s)$ is defined as the ratio of the Laplace transform of the output signal $Y(s)$ to the Laplace transform of the input signal $X(s)$, under the condition that all initial conditions are zero:

$$H(s) = \frac{Y(s)}{X(s)} \quad (1.1.12.1)$$

- For a process driven by an input forcing function $g(t)$ with Laplace transform $G(s)$, the transfer function is written as:

$$H(s) = \frac{Y(s)}{G(s)} \quad (1.1.12.2)$$

where $G(s)$ denotes the input function and $Y(s)$ is the process output response evaluated under relaxed (zero) initial conditions.

- In the time domain, this corresponds to the convolution of the input signal with the system impulse response $h(t) = \mathcal{L}^{-1} \{H(s)\}$:

$$y(t) = h(t) * g(t) \quad (1.1.12.3)$$

1.1.13 Identity Condition for Linear Combinations

- For any independent variables or functions x and y , if the relation

$$ax + by = 0 \quad (1.1.13.1)$$

holds identically for all x and y , then the coefficients must vanish simultaneously:

$$a = 0 \quad \text{and} \quad b = 0 \quad (1.1.13.2)$$

- Similarly, if an expression $C_1 f_1(t) + C_2 f_2(t) = 0$ holds for arbitrary constants C_1 and C_2 , each coefficient must vanish independently:

$$f_1(t) = 0 \quad \text{and} \quad f_2(t) = 0 \quad (1.1.13.3)$$

For distinct exponential functions $e^{r_1 t}$ and $e^{r_2 t}$ ($r_1 \neq r_2$), this implies linear independence across all t .

1.1.14 Continuity and Differentiability in Functions

- Continuity at a Point: A real-valued function $f(x)$ is continuous at $x = x_0$ if and only if the left-hand limit, the right-hand limit, and the value of the function

coincide:

$$\lim_{x \rightarrow x_0^-} f(x) = \lim_{x \rightarrow x_0^+} f(x) = f(x_0) \quad (1.1.14.1)$$

- **Differentiability and Slope Matching:** A function $f(x)$ is differentiable at $x = x_0$ if the left-hand derivative equals the right-hand derivative:

$$f'(x_0^-) = \lim_{h \rightarrow 0^-} \frac{f(x_0 + h) - f(x_0)}{h} = \lim_{h \rightarrow 0^+} \frac{f(x_0 + h) - f(x_0)}{h} = f'(x_0^+) \quad (1.1.14.2)$$

For piecewise functions defined by differentiable curves on each side of the boundary x_0 , this requires matching boundary slopes:

$$f'(x_0^-) = f'(x_0^+) \quad (1.1.14.3)$$

- **Differentiability Implies Continuity:** If $f(x)$ is differentiable at $x = x_0$, it must necessarily be continuous at $x = x_0$:

$$\lim_{x \rightarrow x_0} [f(x) - f(x_0)] = \lim_{x \rightarrow x_0} \left[\frac{f(x) - f(x_0)}{x - x_0} (x - x_0) \right] = f'(x_0) \cdot 0 = 0 \quad (1.1.14.4)$$

Hence, continuity at $x = x_0$ is a necessary prerequisite for differentiability.

1.1.15 Piecewise Boundary Conditions at $x = a$

- Consider a function $f(x)$ defined by differentiable branches $f_1(x)$ and $f_2(x)$ across a boundary point $x = a$:

$$f(x) = \begin{cases} f_1(x), & x \leq a \\ f_2(x), & x > a \end{cases} \quad (1.1.15.1)$$

For $f(x)$ to be continuous at $x = a$, the one-sided limits must equal the boundary value:

$$\lim_{x \rightarrow a^-} f_1(x) = \lim_{x \rightarrow a^+} f_2(x) \implies f_1(a) = f_2(a) \quad (1.1.15.2)$$

For $f(x)$ to be differentiable at $x = a$, the one-sided derivatives at the boundary must coincide:

$$f'_1(a) = f'_2(a) \quad (1.1.15.3)$$

1.1.16 Representation of Linear Balance Systems as Matrix Equations

- A general system of m linear equations in n variables $x_1, x_2, \dots, x_n$:

$$a_{i1}x_1 + a_{i2}x_2 + \dots + a_{in}x_n = b_i, \quad i = 1, 2, \dots, m \quad (1.1.16.1)$$

can be expressed in matrix-vector form:

$$\mathbf{Ax} = \mathbf{b} \quad (1.1.16.2)$$

where $\mathbf{A} \in \mathbb{R}^{m \times n}$ is the coefficient matrix, $\mathbf{x} \in \mathbb{R}^n$ is the state vector, and $\mathbf{b} \in \mathbb{R}^m$

is the constant vector:

$$\mathbf{A} = \begin{pmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & \dots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \dots & a_{mn} \end{pmatrix}, \quad \mathbf{x} = \begin{pmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{pmatrix}, \quad \mathbf{b} = \begin{pmatrix} b_1 \\ b_2 \\ \vdots \\ b_m \end{pmatrix} \quad (1.1.16.3)$$

- In multi-component conservation and stoichiometric balances, the conservation of elemental species across reactant and product streams is represented by:

$$\mathbf{A}_{\text{reactants}} \mathbf{x}_{\text{reactants}} = \mathbf{A}_{\text{products}} \mathbf{x}_{\text{products}} \quad (1.1.16.4)$$

Combining both sides into a single stoichiometric matrix yields the homogeneous system:

$$[\mathbf{A}_{\text{reactants}} \quad -\mathbf{A}_{\text{products}}] \mathbf{x} = \mathbf{0} \quad (1.1.16.5)$$

When boundary parameters or specific yields fix certain degrees of freedom (such as basis mole values and yield coefficients), the homogeneous system reduces to a deterministic non-homogeneous matrix equation $\mathbf{A}\mathbf{x} = \mathbf{b}$ that can be solved directly via row operations or symbolic matrix inversion.

1.1.17 Solution of First-Order Recurrence Relations via Z-Transform

- The bilateral Z-transform of a discrete-time sequence $x(n)$ is defined as:

$$X(z) = \mathcal{Z}(x(n)) = \sum_{n=-\infty}^{\infty} x(n) z^{-n} \quad (1.1.17.1)$$

over the Region of Convergence (ROC) where the series converges absolutely.

- Discrete Unit Step Sequence: The discrete unit step function $u(n)$ is defined as:

$$u(n) = \begin{cases} 1, & n \geq 0 \\ 0, & n < 0 \end{cases} \quad (1.1.17.2)$$

Its forward Z-transform is given by:

$$\mathcal{Z}(u(n)) = \sum_{n=0}^{\infty} z^{-n} = \frac{1}{1 - z^{-1}}, \quad |z| > 1 \quad (1.1.17.3)$$

- Time-Shift Property: Shifting a sequence by an integer delay k corresponds to multiplication by z^{-k} in the Z-domain:

$$\mathcal{Z}(x(n - k)) = z^{-k} X(z) \quad (1.1.17.4)$$

Consequently, delaying the discrete unit step by one sample yields:

$$\mathcal{Z}(u(n - 1)) = \frac{z^{-1}}{1 - z^{-1}}, \quad |z| > 1 \quad (1.1.17.5)$$

- Standard Inverse Transform Pair: For any scalar constant α :

$$\mathcal{Z}^{-1} \left(\frac{1}{1 - \alpha z^{-1}} \right) = \alpha^n u(n), \quad |z| > |\alpha| \quad (1.1.17.6)$$

1.1.18 Relation Between Eigenvalues and Trace of a Matrix

- Let $\mathbf{A} \in \mathbb{R}^{n \times n}$ be a square matrix with eigenvalues $\lambda_1, \lambda_2, \dots, \lambda_n$. The characteristic polynomial of $\mathbf{A}$ is defined by:

$$P(\lambda) = |\lambda \mathbf{I} - \mathbf{A}| \quad (1.1.18.1)$$

- Factoring the polynomial in terms of its n characteristic roots:

$$P(\lambda) = \prod_{i=1}^n (\lambda - \lambda_i) = \lambda^n - \left(\sum_{i=1}^n \lambda_i \right) \lambda^{n-1} + \dots + (-1)^n \prod_{i=1}^n \lambda_i \quad (1.1.18.2)$$

- Evaluating the determinant $|\lambda \mathbf{I} - \mathbf{A}|$ using cofactor expansion, the only term that contributes to λ^n and λ^{n-1} is the product of the principal diagonal entries:

$$\prod_{i=1}^n (\lambda - a_{ii}) = \lambda^n - \left(\sum_{i=1}^n a_{ii} \right) \lambda^{n-1} + \dots \quad (1.1.18.3)$$

where a_{ii} denotes the diagonal elements of $\mathbf{A}$.

- Equating the coefficient of λ^{n-1} from (1.1.18.2) and (1.1.18.3):

$$\sum_{i=1}^n \lambda_i = \sum_{i=1}^n a_{ii} = \text{tr}(\mathbf{A}) \quad (1.1.18.4)$$

Hence, the sum of all eigenvalues of a square matrix is always equal to its trace.

B. Problems

- 1.2.1 An electricity utility company charges ₹7 per kWh (kilo watt-hour). If a 40-watt desk light is left on for 10 hours each night for 180 days, what would be the cost of energy consumption? If the desk light is on for 2 more hours each night for the 180 days, what would be the percentage-increase in the cost of energy consumption?

(GATE CE2 2025)

- a) ₹604.8; 10%
- b) ₹504; 20%
- c) ₹604.8; 12%
- d) ₹720; 15%

Solution: The initial energy consumption is

$$E_1 = P \times t_1 \times N \quad (1.2.1.1)$$

$$= 0.04 \text{ kW} \times 10 \text{ hours/night} \times 180 \text{ days} \quad (1.2.1.2)$$

$$= 72 \text{ kWh} \quad (1.2.1.3)$$

The initial cost at ₹7 per kWh is

$$\text{Initial Cost} = 72 \text{ kWh} \times ₹7/\text{kWh} \quad (1.2.1.4)$$

$$= ₹504 \quad (1.2.1.5)$$

With $t_2 = 12$ hours/night, the new energy consumption and cost are

$$E_2 = 0.04 \text{ kW} \times 12 \text{ hours/night} \times 180 \text{ days} \quad (1.2.1.6)$$

$$= 86.4 \text{ kWh}, \quad (1.2.1.7)$$

$$\text{New Cost} = 86.4 \text{ kWh} \times ₹7/\text{kWh} \quad (1.2.1.8)$$

$$= ₹604.8 \quad (1.2.1.9)$$

Therefore, the percentage increase is

$$\text{Percentage Increase} = \frac{604.8 - 504}{504} \times 100 \quad (1.2.1.10)$$

$$= 20\%. \quad (1.2.1.11)$$

Thus, option (B) is correct.

codes/ce2q3/ce2q3.py

- 1.2.2 A bag contains Violet (V), Yellow (Y), Red (R), and Green (G) balls. On counting them, the following results are obtained:

- (i) The sum of Yellow balls and twice the number of Violet balls is 50.
- (ii) The sum of Violet and Green balls is 50.
- (iii) The sum of Yellow and Red balls is 50.
- (iv) The sum of Violet and twice the number of Red balls is 50.

Which one of the following correctly represents the balls in the bag?

(GATE CE2 2025)

- a) $V = 10\%$, $R = 20\%$, $Y = 30\%$, $G = 40\%$
- b) $V = 40\%$, $R = 10\%$, $Y = 20\%$, $G = 30\%$
- c) $V = 30\%$, $R = 40\%$, $Y = 10\%$, $G = 10\%$
- d) $V = 20\%$, $R = 30\%$, $Y = 40\%$, $G = 10\%$

Solution: Representing the ball counts as $\mathbf{x} = (V \ Y \ R \ G)^\top$, the given conditions form the

matrix equation $\mathbf{Ax} = \mathbf{b}$:

$$\begin{pmatrix} 2 & 1 & 0 & 0 \\ 1 & 0 & 0 & 1 \\ 0 & 1 & 1 & 0 \\ 1 & 0 & 2 & 0 \end{pmatrix} \begin{pmatrix} V \\ Y \\ R \\ G \end{pmatrix} = \begin{pmatrix} 50 \\ 50 \\ 50 \\ 50 \end{pmatrix} \quad (1.2.2.1)$$

Solving the system gives $V = 10$, $Y = 30$, $R = 20$, and $G = 40$. The total number of balls is $T = 100$, so

$$V = 10\%, \quad Y = 30\%, \quad R = 20\%, \quad G = 40\%. \quad (1.2.2.2)$$

Thus, option (A) is correct.

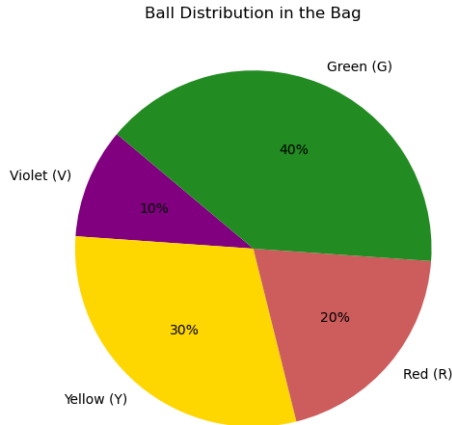


Fig. 1.2.2.1. Pie chart representing ball distribution.

codes/ce2q5/ce2q5.py

- 1.2.3 Consider a five-digit number $PQRST$ that has distinct digits P , Q , R , S , and T , and satisfies the following conditions: $P < Q$, $S > P > T$, and $R < T$. If integers 1 through 5 are used to construct such a number, the value of P is:

(GATE CE2 2025)

- a) 1
- b) 2
- c) 3
- d) 4

Solution: Combining the inequalities gives

$$R < T < P < Q, \quad T < P < S. \quad (1.2.3.1)$$

Since $R < T < P$, the digit P must be greater than at least two other distinct digits, namely 1 and 2. Hence $P \neq 1, 2$.

Also, $P < Q$ and $P < S$, so P cannot be the maximum digit 5. Therefore, the only remaining possibility is $P = 3$.

For $P = 3$, the smaller digits $\{1, 2\}$ can be assigned to R and T , while the larger digits $\{4, 5\}$ can be assigned to Q and S . Hence all the inequalities are satisfied.

Thus, option (C) is correct.

codes/ce2q8/ce2q8.py

- 1.2.4 A business person buys potatoes of two different varieties P and Q , mixes them in a certain ratio and sells them at ₹192 per kg. The cost of variety P is ₹800 for 5 kg. The cost of variety Q is ₹800 for 4 kg. If the person gets 8% profit, what is the $P : Q$ ratio (by weight)? (GATE CE2 2025)

- a) 5 : 4
b) 3 : 4
c) 3 : 2
d) 1 : 1

Solution: Let the profit percentage be $p = 0.08$. Since $SP = CP(1 + p)$, the cost price of the mixture is

$$CP = \frac{SP}{1 + p} \quad (1.2.4.1)$$

$$= \frac{192}{1.08} = \frac{1600}{9} \text{ ₹/kg.} \quad (1.2.4.2)$$

The unit costs of the two varieties are

$$C_P = \frac{800}{5} = 160 \text{ ₹/kg,} \quad (1.2.4.3)$$

$$C_Q = \frac{800}{4} = 200 \text{ ₹/kg.} \quad (1.2.4.4)$$

Let the weights mixed be W_P and W_Q . Equating the total cost of the mixture to the component costs gives

$$W_P C_P + W_Q C_Q = (W_P + W_Q) CP \quad (1.2.4.5)$$

$$W_P (CP - C_P) = W_Q (C_Q - CP). \quad (1.2.4.6)$$

Hence,

$$\frac{W_P}{W_Q} = \frac{C_Q - CP}{CP - C_P} \quad (1.2.4.7)$$

$$= \frac{200 - \frac{1600}{9}}{\frac{1600}{9} - 160} \quad (1.2.4.8)$$

$$= \frac{\frac{200}{9}}{\frac{160}{9}} = \frac{5}{4}. \quad (1.2.4.9)$$

Thus, the required ratio is 5 : 4, making option (A) correct.

codes/ce2q9/ce2q9.py

- 1.2.5 Three villages P , Q , and R are located in such a way that the distance $PQ = 13$ km, $QR = 14$ km, and $RP = 15$ km. A straight road joins Q and R . It is proposed to connect P to this road QR by constructing another road. What is the minimum possible length (in km) of this connecting road?

(GATE CE2 2025)

- a) 10.5
b) 11.0
c) 12.0
d) 12.5

Solution: The minimum distance from vertex P to line QR is the altitude h_a . The semi-

perimeter of triangle PQR is

$$s = \frac{13 + 14 + 15}{2} = 21. \quad (1.2.5.1)$$

Using Heron's formula, the area is

$$\Delta = \sqrt{21(21 - 14)(21 - 15)(21 - 13)} \quad (1.2.5.2)$$

$$= \sqrt{21 \times 7 \times 6 \times 8} \quad (1.2.5.3)$$

$$= 84 \text{ km}^2. \quad (1.2.5.4)$$

Taking $QR = 14$ km as the base,

$$h_a = \frac{2\Delta}{14} = \frac{2 \times 84}{14} = 12 \text{ km}. \quad (1.2.5.5)$$

Thus, option (C) is correct.

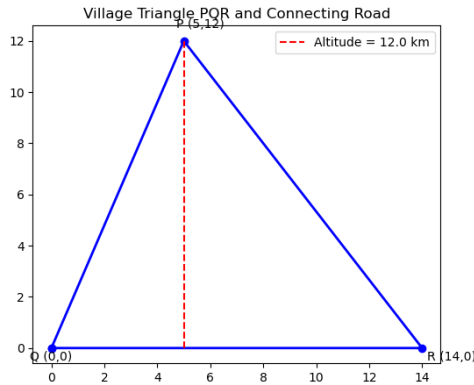


Fig. 1.2.5.1. Triangle PQR geometry and altitude from P to QR .

codes/ce2q10/ce2q10.py

1.2.6 For the matrix $[\mathbf{A}]$ given below, the transpose is:

$$[\mathbf{A}] = \begin{pmatrix} 2 & 3 & 4 \\ 1 & 4 & 5 \\ 4 & 3 & 2 \end{pmatrix} \quad (1.2.6.1)$$

(GATE CE2 2025)

- a) $\begin{pmatrix} 2 & 1 & 4 \\ 3 & 4 & 3 \\ 4 & 5 & 2 \end{pmatrix}$
- b) $\begin{pmatrix} 4 & 3 & 2 \\ 5 & 4 & 1 \\ 2 & 3 & 4 \end{pmatrix}$
- c) $\begin{pmatrix} 4 & 2 & 3 \\ 5 & 1 & 4 \\ 2 & 4 & 3 \end{pmatrix}$

d) $\begin{pmatrix} 2 & 3 & 4 \\ 1 & 4 & 5 \\ 4 & 3 & 2 \end{pmatrix}$

Solution: The transpose operation interchanges rows and columns. Therefore,

$$[\mathbf{A}^T]_{ij} = [\mathbf{A}]_{ji}, \quad (1.2.6.2)$$

$$\mathbf{A}^T = \begin{pmatrix} 2 & 1 & 4 \\ 3 & 4 & 3 \\ 4 & 5 & 2 \end{pmatrix}. \quad (1.2.6.3)$$

Thus, option (A) is correct.

codes/ce2q11/ce2q11.py

1.2.7 Integration of $\ln(x)$ with respect to x , i.e., $\int \ln(x) dx =$ _____ (GATE CE2 2025)

- a) $x \ln(x) - x + C$
- b) $x - \ln(x) + C$
- c) $x \ln(x) + x + C$
- d) $\ln(x) - x + C$

Solution: Using integration by parts with $u = \ln(x)$ and $dv = dx$,

$$\int \ln(x) dx = x \ln(x) - \int x \cdot \frac{1}{x} dx \quad (1.2.7.1)$$

$$= x \ln(x) - \int 1 dx \quad (1.2.7.2)$$

$$= x \ln(x) - x + C. \quad (1.2.7.3)$$

Thus, option (A) is correct.

codes/ce2q12/ce2q12.py

1.2.8 Consider the pin-jointed truss shown in the figure. Influence line is drawn for the axial force in the member $G - I$, when a unit load travels on the bottom chord of the truss. Identify the CORRECT influence line from the options provided.

(GATE CE2 2025)

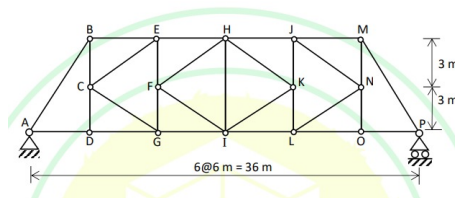


Fig. 1.2.8.1. Influence line diagram for axial force in member $G - I$.

Solution: Passing a vertical section through the bottom chord member $G - I$ and taking moments about node H , the influence-line ordinate reaches its peak at nodes G and I . For a truss of height $h = 3$ m and panel length $L = 6$ m, the peak tensile ordinate is 1.33. Thus, option (B) is correct.

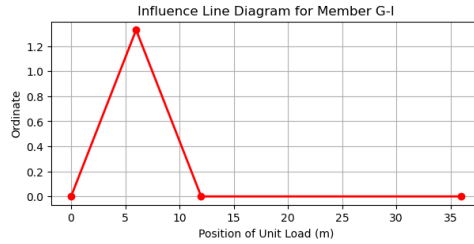


Fig. 1.2.8.2. Influence line diagram for axial force in member $G - I$.

codes/ce2q14/ce2q14.py

1.2.9 Consider a velocity vector $\mathbf{V}$ in (x, y, z) coordinates given by $\mathbf{V} = u\mathbf{x} + v\mathbf{y}$. Pick one or more CORRECT statement(s) from the choices given below.

(GATE CE2 2025)

- a) z -component of Curl of velocity: $\nabla \times \mathbf{V} = \left(\frac{\partial v}{\partial x} - \frac{\partial u}{\partial y} \right) \mathbf{z}$
- b) z -component of Curl of velocity: $\nabla \times \mathbf{V} = \left(\frac{\partial u}{\partial x} - \frac{\partial v}{\partial y} \right) \mathbf{z}$
- c) Divergence of velocity: $\nabla \cdot \mathbf{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}$
- d) Divergence of velocity: $\nabla \cdot \mathbf{V} = \frac{\partial u}{\partial y} + \frac{\partial v}{\partial x}$

Solution: For a two-dimensional velocity field $\mathbf{V} = u\mathbf{x} + v\mathbf{y}$, the z -component of curl is

$$(\nabla \times \mathbf{V})_z = \frac{\partial v}{\partial x} - \frac{\partial u}{\partial y}. \quad (1.2.9.1)$$

Thus, statement (A) is correct.

The divergence is

$$\nabla \cdot \mathbf{V} = \frac{\partial u}{\partial x} + \frac{\partial v}{\partial y}. \quad (1.2.9.2)$$

Thus, statement (C) is correct.

Therefore, the correct statements are (A) and (C).

codes/ce2q21/ce2q21.py

1.2.10 Given that A and B are not null sets, which of the following statements regarding probability is/are CORRECT?

(GATE CE2 2025)

- a) $\Pr(A \cap B) = \Pr(A)\Pr(B)$ if A and B are mutually exclusive.
- b) Conditional probability, $\Pr(A | B) = 1$ if $B \subset A$.
- c) $\Pr(A \cup B) = \Pr(A) + \Pr(B)$, if A and B are mutually exclusive.
- d) $\Pr(A \cap B) = 0$, if A and B are independent.

Solution: Since $B \subset A$,

$$A \cap B = B, \quad (1.2.10.1)$$

$$\Pr(A | B) = \frac{\Pr(A \cap B)}{\Pr(B)} = \frac{\Pr(B)}{\Pr(B)} = 1. \quad (1.2.10.2)$$

Hence, statement (B) is correct.

For mutually exclusive events,

$$\Pr(A \cap B) = 0, \quad (1.2.10.3)$$

$$\Pr(A \cup B) = \Pr(A) + \Pr(B). \quad (1.2.10.4)$$

Hence, statement (C) is correct. Statements (A) and (D) are incorrect for non-null sets. Therefore, the correct statements are (B) and (C).

`codes/ce2q22/ce2q22.py`

1.2.11 Pick the CORRECT solution for the following differential equation

$$\frac{dy}{dx} = e^{x-y}$$

(GATE CE2 2025)

- a) $y = \ln(e^x + \text{Constant})$
- b) $\ln(y) = x + \text{Constant}$
- c) $\ln(y) = \ln(e^x) + \text{Constant}$
- d) $y = x + \text{Constant}$

Solution: The given differential equation is nonlinear:

$$\frac{dy}{dx} = e^{x-y} = e^x e^{-y}$$

To use Laplace transformation, we first convert it into a linear differential equation by introducing the substitution

$$u = e^y$$

Then,

$$\frac{du}{dx} = e^y \frac{dy}{dx}$$

Using the given equation,

$$\frac{du}{dx} = e^y \cdot e^{x-y} = e^x$$

Thus, the transformed problem reduces to

$$\frac{du}{dx} = e^x$$

Now take Laplace transforms on both sides with respect to x . Let

$$\mathcal{L}\{u(x)\} = U(s)$$

Then,

$$\mathcal{L}\left\{\frac{du}{dx}\right\} = sU(s) - u(0)$$

and

$$\mathcal{L}\{e^x\} = \frac{1}{s-1}$$

Hence,

$$sU(s) - u(0) = \frac{1}{s-1}$$

Therefore,

$$U(s) = \frac{1}{s(s-1)} + \frac{u(0)}{s}$$

Using partial fractions,

$$\frac{1}{s(s-1)} = -\frac{1}{s} + \frac{1}{s-1}$$

So,

$$U(s) = -\frac{1}{s} + \frac{1}{s-1} + \frac{u(0)}{s}$$

$$U(s) = \frac{u(0)-1}{s} + \frac{1}{s-1}$$

Taking inverse Laplace transform,

$$u(x) = u(0) - 1 + e^x$$

Since $u = e^y$, we get

$$e^y = e^x + C$$

where $C = u(0) - 1$ is an arbitrary constant.

Taking natural logarithm,

$$y = \ln(e^x + C)$$

Hence, the correct solution is

$$y = \ln(e^x + \text{Constant})$$

Therefore, the correct option is

$$(A)$$

codes/ce2q36/ce2q36.py

1.2.12 The figure shows a propped cantilever with uniform flexural rigidity

$$EI$$

(in N.m

$$2$$

) and subjected to a moment

$$M$$

(in N.m). Consider forces and displacements in the upward direction as positive. Find the upward reaction at the propped support

$$B$$

(in N) when this support settles by

$$(-\Delta)$$

, given in metres.

(GATE CE2 2025)

a)

$$\frac{3M}{2L} - \frac{6EI\Delta}{L^3}$$

b)

$$\frac{8M}{3L} - \frac{2EI\Delta}{L^3}$$

c)

$$\frac{3M}{2L} - \frac{3EI\Delta}{L^3}$$

d)

$$\frac{M}{L} - \frac{8EI\Delta}{L^3}$$

Solution: Treat the reaction at the prop

B

as the redundant

R_B

. The basic determinate structure is a cantilever fixed at

A

and free at

B

. Using compatibility, the net vertical deflection at

B

must equal the imposed support settlement. Since upward displacement is taken positive and the support settles downward by

$$(-\Delta)$$

,

$$\delta_B = -\Delta$$

The end deflection at

B

due to an end moment

M

on a cantilever of length

L

is

$$\delta_M = -\frac{ML^2}{2EI}$$

The end deflection at

B

due to an upward redundant force

R_B

applied at the free end of the cantilever is

$$\delta_R = \frac{R_B L^3}{3EI}$$

By superposition,

$$\delta_M + \delta_R = -\Delta$$

Hence,

$$-\frac{ML^2}{2EI} + \frac{R_B L^3}{3EI} = -\Delta$$

Multiplying throughout by

$$3EI/L^3$$

,

$$-\frac{3M}{2L} + R_B = -\frac{3EI\Delta}{L^3}$$

Therefore,

$$R_B = \frac{3M}{2L} - \frac{3EI\Delta}{L^3}$$

Thus, the upward reaction at support

B

is

$$\boxed{\frac{3M}{2L} - \frac{3EI\Delta}{L^3}}$$

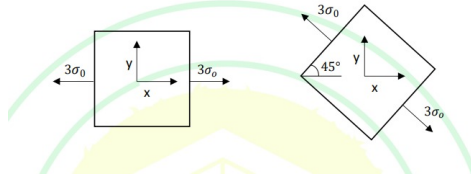
Hence, the correct option is

$\boxed{(C)}$

`codes/ce2q38/ce2q38.py`

- 1.2.13 Let the state of stress at a point in a body be the difference of two plane states of stress shown in the figure. Consider all the possible planes perpendicular to the x - y plane and passing through that point. The magnitude of the maximum compressive stress on any such plane is $k\sigma_0$, where k is equal to _____ (round off to one decimal place).

(GATE CE2 2025)



Solution: The required stress tensor is the difference of the two given plane stress states. For the first figure, the stress is a uniaxial tensile stress of magnitude $3\sigma_0$ along the x -direction. Hence,

$$[\sigma^{(1)}] = \begin{bmatrix} 3\sigma_0 & 0 \\ 0 & 0 \end{bmatrix}$$

For the second figure, the element is rotated by 45° , and the uniaxial tensile stress $3\sigma_0$ acts along that rotated axis. In its local principal coordinates,

$$[\sigma'] = \begin{bmatrix} 3\sigma_0 & 0 \\ 0 & 0 \end{bmatrix}$$

Transforming to the global x - y coordinates for $\theta = 45^\circ$,

$$[\sigma^{(2)}] = \begin{bmatrix} 3\sigma_0 \cos^2 45^\circ & 3\sigma_0 \sin 45^\circ \cos 45^\circ \\ 3\sigma_0 \sin 45^\circ \cos 45^\circ & 3\sigma_0 \sin^2 45^\circ \end{bmatrix}$$

Using

$$\cos^2 45^\circ = \sin^2 45^\circ = \frac{1}{2}, \quad \sin 45^\circ \cos 45^\circ = \frac{1}{2},$$

we get

$$[\sigma^{(2)}] = \begin{bmatrix} \frac{3\sigma_0}{2} & \frac{3\sigma_0}{2} \\ \frac{3\sigma_0}{2} & \frac{3\sigma_0}{2} \end{bmatrix}$$

Hence, the required resultant stress tensor is

$$[\sigma] = [\sigma^{(1)}] - [\sigma^{(2)}] = \begin{bmatrix} \frac{3\sigma_0}{2} & -\frac{3\sigma_0}{2} \\ -\frac{3\sigma_0}{2} & -\frac{3\sigma_0}{2} \end{bmatrix}$$

The principal stresses are the eigenvalues of this matrix. Therefore,

$$\det \left(\begin{bmatrix} \frac{3\sigma_0}{2} - \lambda & -\frac{3\sigma_0}{2} \\ -\frac{3\sigma_0}{2} & -\frac{3\sigma_0}{2} - \lambda \end{bmatrix} \right) = 0$$

Expanding,

$$\left(\frac{3\sigma_0}{2} - \lambda \right) \left(-\frac{3\sigma_0}{2} - \lambda \right) - \left(\frac{3\sigma_0}{2} \right)^2 = 0$$

$$\lambda^2 - \frac{9\sigma_0^2}{2} = 0$$

Hence,

$$\lambda = \pm \frac{3\sigma_0}{\sqrt{2}}$$

The maximum compressive stress magnitude is therefore

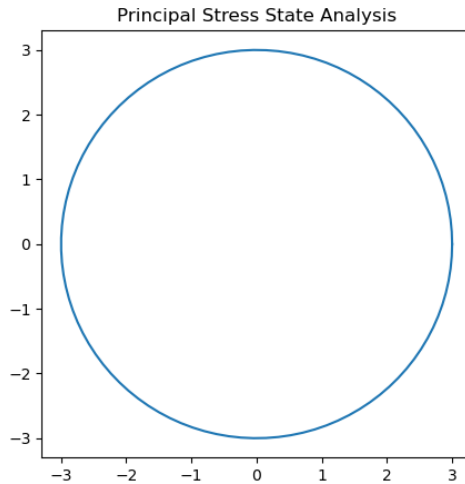
$$\frac{3\sigma_0}{\sqrt{2}} = 2.121\sigma_0$$

Thus,

$$k = 2.1$$

2.1

codes/ce2q39/ce2q39.py



1.2.14 Consider the function given below and pick one or more CORRECT statement(s) from the following choices.

$$f(x) = x^3 - \frac{15}{2}x^2 + 18x + 20$$

(GATE CE2 2025)

- a) $f(x)$ has a local minimum at $x = 3$.
- b) $f(x)$ has a local maximum at $x = 3$.
- c) $f(x)$ has a local minimum at $x = 2$.
- d) $f(x)$ has a local maximum at $x = 2$.

Solution: To determine the local extrema, we differentiate the function.

$$f'(x) = 3x^2 - 15x + 18$$

Factorizing,

$$f'(x) = 3(x^2 - 5x + 6) = 3(x - 2)(x - 3)$$

Thus, the critical points are

$$x = 2, \quad x = 3$$

Now compute the second derivative:

$$f''(x) = 6x - 15$$

At $x = 2$,

$$f''(2) = 12 - 15 = -3 < 0$$

Hence, $f(x)$ has a local maximum at $x = 2$.

At $x = 3$,

$$f''(3) = 18 - 15 = 3 > 0$$

Hence, $f(x)$ has a local minimum at $x = 3$.

Therefore, the correct statements are

(A) and (D)

codes/ce2q44/ce2q44.py

1.2.15 Pick the CORRECT eigenvalue(s) of the matrix $[A]$ from the following choices.

$$[A] = \begin{bmatrix} 6 & 8 \\ 4 & 2 \end{bmatrix}$$

(GATE CE2 2025)

- a) 10
- b) 4
- c) -2
- d) -10

Solution: The eigenvalues are obtained from the characteristic equation

$$\det(A - \lambda I) = 0$$

Therefore,

$$\det \begin{bmatrix} 6 - \lambda & 8 \\ 4 & 2 - \lambda \end{bmatrix} = 0$$

Expanding,

$$(6 - \lambda)(2 - \lambda) - 32 = 0$$

$$12 - 8\lambda + \lambda^2 - 32 = 0$$

$$\lambda^2 - 8\lambda - 20 = 0$$

Factoring,

$$(\lambda - 10)(\lambda + 2) = 0$$

Hence, the eigenvalues are

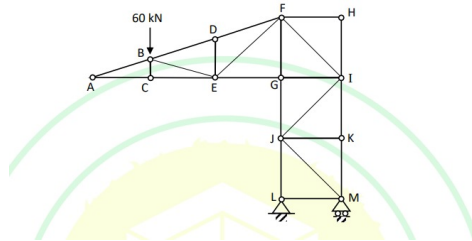
$$\lambda = 10, \quad \lambda = -2$$

So, the correct options are

(A) and (C)

codes/ce2q45/ce2q45.py

- 1.2.16 In the pin-jointed truss shown in the figure, the members that carry zero force are identified. Which of the following options is/are zero-force members?



(GATE CE2 2025)

- a) BC
- b) EG
- c) FI
- d) JK

Solution: We apply the standard zero-force member rules.

At joint H , only two non-collinear members FH and HI meet, and there is no external load or support reaction at that joint. Therefore,

$$FH = 0, \quad HI = 0$$

Now consider joint K . Since $HI = 0$, the joint $I-K-M$ vertical line does not transmit any force from above through HI . At joint K , three members meet: IK and KM are collinear, and JK is non-collinear. There is no external load at joint K . Hence, by the zero-force member rule,

$$JK = 0$$

Thus, among the listed options, JK is definitely a zero-force member.

The other listed members BC , EG , and FI are connected to joints that are either loaded, connected to more than three members, or do not satisfy the zero-force member criterion directly. Hence they are not zero-force members.

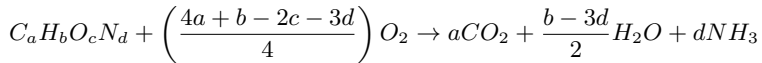
Therefore, the correct option is

(D)

codes/ce2q46/ce2q46.py

- 1.2.17 A compound has a general formula $C_aH_bO_cN_d$ and molecular weight 187. A 935 mg/l solution of the compound is prepared in distilled deionized water. The Total Organic Carbon (TOC) is measured as 360 mg/l (as C). The Chemical Oxygen Demand (COD) and the Total Kjeldahl Nitrogen (TKN) are determined as 600 mg/l (as O_2) and 140 mg/l (as N), respectively.

As per the reaction



which of the following options is/are CORRECT?

Atomic weights: $C = 12$, $H = 1$, $O = 16$, $N = 14$

(GATE CE2 2025)

- a) $a = 6$
- b) $b = 7$
- c) $c = 5$
- d) $d = 3$

Solution: The compound concentration is 935 mg/l and molecular weight is 187. Therefore, the molar concentration is

$$\frac{935}{187} = 5 \text{ mmol/l}$$

So the solution contains 5×10^{-3} mol/l of the compound.

From TOC,

$$(5 \times 10^{-3}) \times 12a = 360 \times 10^{-3}$$

$$0.06a = 0.36$$

$$a = 6$$

From TKN,

$$(5 \times 10^{-3}) \times 14d = 140 \times 10^{-3}$$

$$0.07d = 0.14$$

$$d = 2$$

Hence option (D) is false.

Now use molecular weight:

$$12a + b + 16c + 14d = 187$$

Substituting $a = 6$ and $d = 2$,

$$72 + b + 16c + 28 = 187$$

$$b + 16c = 87$$

From COD data, oxygen required is 600 mg/l = 0.6 g/l.

For 5×10^{-3} mol/l of compound, oxygen consumed per mole of compound is

$$\frac{0.6}{5 \times 10^{-3}} = 120 \text{ g/mol of } O_2$$

Since molecular weight of O_2 is 32,

$$\text{moles of } O_2 = \frac{120}{32} = 3.75$$

From the given reaction,

$$\frac{4a + b - 2c - 3d}{4} = 3.75$$

Substituting $a = 6$ and $d = 2$,

$$\frac{24 + b - 2c - 6}{4} = 3.75$$

$$18 + b - 2c = 15$$

$$b - 2c = -3$$

Now solve the simultaneous equations

$$b + 16c = 87$$

$$b - 2c = -3$$

Subtracting,

$$18c = 90$$

$$c = 5$$

Then,

$$b - 2(5) = -3$$

$$b = 7$$

Thus,

$$a = 6, \quad b = 7, \quad c = 5, \quad d = 2$$

Therefore, the correct options are

(A), (B) and (C)

codes/ce2q49/ce2q49.py

- 1.2.18 Consider a discrete random variable X whose probabilities are given below. The standard deviation of the random variable is _____ (round off to one decimal place).

x_i	1	2	4	8
$P(X = x_i)$	0.3	0.1	0.3	0.3

Solution: The mean is

(GATE CE2 2025)

$$\mu = E[X] = \sum x_i p_i$$

$$\mu = 1(0.3) + 2(0.1) + 4(0.3) + 8(0.3)$$

$$\mu = 0.3 + 0.2 + 1.2 + 2.4 = 4.1$$

Now,

$$E[X^2] = 1^2(0.3) + 2^2(0.1) + 4^2(0.3) + 8^2(0.3)$$

$$E[X^2] = 0.3 + 0.4 + 4.8 + 19.2 = 24.7$$

Hence the variance is

$$\sigma^2 = E[X^2] - \mu^2$$

$$\sigma^2 = 24.7 - (4.1)^2 = 24.7 - 16.81 = 7.89$$

Therefore, the standard deviation is

$$\sigma = \sqrt{7.89} = 2.81$$

Rounded off to one decimal place,

2.8

codes/ce2q51/ce2q51.py

- 1.2.19 A steel beam supported by three parallel pin-jointed steel rods is shown in the figure. The moment of inertia of the beam is $8 \times 10^7 \text{ mm}^4$. Take modulus of elasticity of steel as 210 GPa. The beam is subjected to uniformly distributed load of 6.25 kN/m, including its self-weight. The axial force (in kN) in the centre rod CD is _____ (round off to one decimal place).

(GATE CE2 2025)

Solution: The beam length is 4 m with three vertical steel rods at B , D , and F . The load is symmetric, and the geometry as well as the support arrangement are also symmetric. Hence,

$$R_B = R_F$$

Total uniformly distributed load on the beam is

$$wL = 6.25 \times 4 = 25 \text{ kN}$$

Therefore,

$$R_B + R_D + R_F = 25$$

or, by symmetry,

$$2R_B + R_D = 25$$

Let the stiffness of side rods AB and EF be k_s , and that of the centre rod CD be k_c . For an axially loaded rod,

$$k = \frac{AE}{L}$$

Since all rods have the same length and material, stiffness is proportional to area. Thus,

$$k \propto d^2$$

For side rods, diameter = 12 mm. For centre rod, diameter = 30 mm.

Hence,

$$\frac{k_c}{k_s} = \left(\frac{30}{12}\right)^2 = 6.25$$

Now, due to symmetry the beam remains horizontal, so vertical deflection at B , D , and F is the same. Therefore, rod forces are proportional to rod stiffness:

$$\frac{R_D}{R_B} = 6.25$$

Thus,

$$R_D = 6.25R_B$$

Substituting into equilibrium:

$$2R_B + 6.25R_B = 25$$

$$8.25R_B = 25$$

$$R_B = 3.03 \text{ kN}$$

Therefore,

$$R_D = 6.25 \times 3.03 = 18.94 \text{ kN}$$

Rounded off to one decimal place,

18.9

codes/ce2q52/ce2q52.py

- 1.2.20 The figure shows a network diagram for a construction project. The activities A, B, C, D, E, and F are represented by arrows and their durations are shown in the figure.

The total float available for the activity E in day(s) is equal to _____ (round off to the nearest integer).

(GATE CE2 2025)

Solution: From the network:

$$A : 1 \rightarrow 2 = 3 \text{ days}$$

$$B : 2 \rightarrow 4 = 5 \text{ days}$$

$$C : 1 \rightarrow 3 = 5 \text{ days}$$

$$D : 3 \rightarrow 4 = 2 \text{ days}$$

$$E : 4 \rightarrow 5 = 4 \text{ days}$$

$$F : 3 \rightarrow 5 = 8 \text{ days}$$

Forward pass:

Earliest event time at node 1:

$$TE_1 = 0$$

At node 2:

$$TE_2 = 0 + 3 = 3$$

At node 3:

$$TE_3 = 0 + 5 = 5$$

At node 4:

$$TE_4 = \max(TE_2 + 5, TE_3 + 2) = \max(3 + 5, 5 + 2) = 8$$

At node 5:

$$TE_5 = \max(TE_4 + 4, TE_3 + 8) = \max(8 + 4, 5 + 8) = 13$$

Backward pass:

$$TL_5 = 13$$

At node 4:

$$TL_4 = TL_5 - 4 = 13 - 4 = 9$$

At node 3:

$$TL_3 = \min(TL_4 - 2, TL_5 - 8) = \min(9 - 2, 13 - 8) = 5$$

At node 2:

$$TL_2 = TL_4 - 5 = 9 - 5 = 4$$

For activity E ($4 \rightarrow 5$), total float is

$$TF_E = TL_5 - TE_4 - t_E$$

$$TF_E = 13 - 8 - 4 = 1$$

Therefore, the total float available for activity E is

$$\boxed{1}$$

codes/ce2q53/ce2q53.py

1.2.21 The shaft of a 6 m wide gate in the figure will fail at a moment of 3924 kN.m about the hinge P . The maximum value of water depth h (in m) that the gate can hold is _____ (round off to the nearest integer).

Note: Density of water = 1000 kg/m³, acceleration due to gravity = 9.81 m/s².

Solution: The inclined gate spans 3 m horizontally and 4 m vertically, so its length is

$$L = \sqrt{3^2 + 4^2} = 5 \text{ m}$$

The width of gate is $b = 6 \text{ m}$. Hence, area of the inclined gate is

$$A = 5 \times 6 = 30 \text{ m}^2$$

Let the water depth above hinge level be h . The top end of the gate is 4 m above the hinge, so the centroid of the inclined rectangular gate is at 2 m above hinge level. Hence its depth below free surface is

$$\bar{y} = h - 2$$

The hydrostatic force on the inclined plane is

$$F = \rho g A \bar{y}$$

$$F = 1000 \times 9.81 \times 30 \times (h - 2)$$

$$F = 294300(h - 2) \text{ N}$$

Since pressure varies linearly from bottom to top, and the top edge is at shallower depth, the resultant acts at a distance from hinge obtained from moment integration along the gate. Take coordinate s along the gate from hinge upward. Vertical elevation of a point is

$$y = \frac{4}{5}s$$

Hence pressure intensity at that point is proportional to

$$p(s) = \rho g \left(h - \frac{4}{5}s \right)$$

Elemental force on strip ds is

$$dF = p(s) b ds = \rho g b \left(h - \frac{4}{5}s \right) ds$$

Moment about hinge:

$$M_P = \int_0^5 s dF$$

$$M_P = \rho g b \int_0^5 s \left(h - \frac{4}{5}s \right) ds$$

$$M_P = \rho g b \left[h \int_0^5 s ds - \frac{4}{5} \int_0^5 s^2 ds \right]$$

Now,

$$\int_0^5 s ds = \frac{25}{2} \quad \text{and} \quad \int_0^5 s^2 ds = \frac{125}{3}$$

Therefore,

$$M_P = \rho g b \left(\frac{25h}{2} - \frac{4}{5} \cdot \frac{125}{3} \right)$$

$$M_P = \rho g b \left(\frac{25h}{2} - \frac{100}{3} \right)$$

Substituting $\rho = 1000$, $g = 9.81$, $b = 6$,

$$M_P = 1000 \times 9.81 \times 6 \left(\frac{25h}{2} - \frac{100}{3} \right)$$

Failure occurs at

$$M_P = 3924 \text{ kN.m} = 3.924 \times 10^6 \text{ N.m}$$

Thus,

$$1000 \times 9.81 \times 6 \left(\frac{25h}{2} - \frac{100}{3} \right) = 3.924 \times 10^6$$

$$58860 \left(\frac{25h}{2} - \frac{100}{3} \right) = 3.924 \times 10^6$$

$$\frac{25h}{2} - \frac{100}{3} = 66.6667$$

$$\frac{25h}{2} = 100$$

$$25h = 200$$

$$h = 8$$

Therefore, the maximum water depth is

8

codes/ce2q60/ce2q60.py

- 1.2.22 On a two-lane highway, a horizontal curve of radius 300 m is provided. The design speed is 80 km/h.

If the longest wheelbase of vehicle expected on this highway is 7 m, then the extra widening required (in m) is _____ (round off to two decimal places).

(GATE CE2 2025)

Solution: Extra widening on a horizontal curve is given by

$$W_e = W_m + W_p$$

where W_m is the mechanical widening and W_p is the psychological widening.

For a two-lane road,

$$W_m = \frac{nl^2}{2R}$$

$$W_p = \frac{V}{9.5\sqrt{R}}$$

Given:

$$n = 2, \quad l = 7 \text{ m}, \quad R = 300 \text{ m}, \quad V = 80 \text{ km/h}$$

Now,

$$W_m = \frac{2 \times 7^2}{2 \times 300} = \frac{49}{300} = 0.1633 \text{ m}$$

Also,

$$W_p = \frac{80}{9.5\sqrt{300}}$$

$$\sqrt{300} = 17.3205$$

$$W_p = \frac{80}{9.5 \times 17.3205} = 0.4862 \text{ m}$$

Hence,

$$W_e = 0.1633 + 0.4862 = 0.6495 \text{ m}$$

Rounded off to two decimal places,

0.65

codes/ce2q64/ce2q64.py

- 1.2.23 If the Fore Bearing of the lines AB and BC are 60° and 122° , respectively, then the interior angle $\angle ABC$ (in degrees) is _____ (round off to the nearest integer).

(GATE CE2 2025)

Solution: The fore bearing of line AB is

$$FB_{AB} = 60^\circ$$

Hence the back bearing of line BA is

$$BB_{BA} = 60^\circ + 180^\circ = 240^\circ$$

The fore bearing of line BC is

$$FB_{BC} = 122^\circ$$

The interior angle at B , that is $\angle ABC$, is the angle between the directions BA and BC . Therefore,

$$\angle ABC = |240^\circ - 122^\circ| = 118^\circ$$

Hence,

118°

